Guesstimate - Netflix

Asked by : **Netflix**

Problem Statement : Estimate the number of planes that are in the air in an hour ?

Clarifying Questions :

Q1: Geography scope?

Ans: India only, domestic flights.

Q2: Flight types—passenger or cargo?

Ans: Passenger flights only. (Exclude cargo, general aviation, military.)

Q3: Include international?

Ans: No. Domestic only.

Q4: Time basis-specific hour or average moment over a day?

Ans: Average moment over a typical day (i.e., expected number airborne at any instant).

Q5: Seasonality/weekday effects?

Ans: Use a **typical (non-peak) day**; ignore holiday spikes.

Approach

I am following up the bottom-up approach using airport throughput.

Equation

Average planes airborne per hour = (# of airports)*(# of planes taking off per hour)

Step 1 - Flights per day

We categorize airports:

- **Busy airports (8 airports)**: assume **3 runways**, avg 1 flight every 5 minutes/runway, 24 hrs

  → 8 × 3 × (60/5) × 24 = **6,912 flights/day**

- **Intermediate airports (30 airports)**: 1 runway, 1 flight every 10 mins, ~75% utilization, 24 hrs

  → 30 × 1 × (60/10) × 0.75 × 24 = **3,240 flights/day**

- **Occasional airports (50 airports)**: 1 runway, 1 flight every 30 mins, ~25% utilization, 24 hrs

  → 50 × 1 × (60/30) × 0.25 × 24 = **600 flights/day**

Total domestic passenger flights/day = 6,912 + 3,240 + 600 = 10,752

Step 2 — Average flight duration assumption

For Indian domestic passenger flights, common trunk routes (Delhi–Mumbai, Mumbai–Bengaluru, Delhi–Pune, etc.) average around **1.5–2.5 hours**. We assume an **average of ~2 hours per flight.**

Step 3 — Total flight-hours per day

= 10,752 flights × 2 hours = **21,504 flight-hours/day**

Step 4 — Average concurrent planes airborne (domestic passenger)

= 21,504 ÷ 24 = **896 planes**

Answer

> At any given moment in India, there are approximately ~900 domestic passenger planes in the air, on average, across a typical day.